“AI-Powered Stock Screening and Analysis Using Multi-Factor Scoring and Steepest-Ascent Hill Climbing”

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***Abstract*—*Investors must evaluate large volumes of stocks before making sound decisions, yet traditional analysis demands manual study of financial statements, historical prices, market trends, and technical indicators across fragmented tools and data sources. This is time-consuming and complex, particularly for beginners. This paper proposes an intelligent, unified system that leverages artificial intelligence to screen and analyze equities. The system integrates fundamental analysis, technical analysis, and news sentiment analysis using Natural Language Processing (NLP) in a single interface, and computes automated scores, rankings, and plain-language explanations for each stock. At the core of the analytical engine is the Steepest-Ascent Hill Climbing algorithm, a local-search technique that iteratively moves toward the neighboring state offering the steepest improvement in score; here it optimizes stock-scoring weights. The system is intended as a scalable, explainable decision-support tool that reduces manual screening effort. It provides analytical insight and is not intended to guarantee returns or serve as financial advice.*

***Keywords*—*Stock Screening, Fundamental Analysis, Technical Analysis, Sentiment Analysis, Natural Language Processing, Hill Climbing, Explainable AI, Decision Support.*

I. Introduction

Stock screening is the process of evaluating a collection of stocks using selected financial, market, and other criteria to identify candidates for further analysis. Traditional screening often requires the investor to inspect financial statements, historical prices, technical indicators, market trends, and financial news separately.

This work addresses the fragmented workflow by proposing a unified AI-powered stock screening and analysis system. Instead of moving between several single-purpose tools, users get fundamental, technical, and sentiment analysis together in one interface.

A. Need for the System

Large volumes of stock information make manual evaluation time-consuming; financial information is spread across different sources and tools; beginners may find multi-factor analysis complex; and automated scoring with plain-language explanations makes results easier to compare.

B. Aim and Objectives

The aim is to develop an intelligent, unified decision-support system for stock screening and analysis by combining fundamental, technical, and sentiment analysis and optimizing the scoring process with Steepest-Ascent Hill Climbing. The objectives are to: provide a unified interface; collect prices, financial statements, trading volume, and news text; process and normalize the data; perform fundamental analysis (ratios, earnings, balance-sheet metrics), technical analysis (Moving Averages, RSI, MACD, price/volume dynamics), and NLP-based sentiment analysis; combine features through an AI/ML engine; optimize scoring weights with hill climbing; and generate scores, rankings, charts, indicators, and AI-generated explanations.

C. Scope

The project focuses on analytical and decision support: screening and multi-factor analysis of equities, integration of three analytical branches, optimization of scoring weights, and presentation of results through a responsive dashboard. It

provides insight rather than guaranteed returns or financial advice.

D. Design Characteristics

The design emphasizes unified analysis rather than disconnected single-purpose tools; multi-branch analysis using fundamental, technical, and sentiment information; optimization of stock-scoring weights using a local-search technique; explainable output through plain-language summaries; and responsive presentation through a web/mobile dashboard.

E. Organization of the Paper

Section II reviews the literature, Section III presents the architecture, Section IV describes the modules and the hill-climbing procedure, Section V covers implementation and results, Section VI discusses design rationale and practical considerations, and Section VII concludes with future work.

II. Literature Survey

The literature identifies several established approaches that form the conceptual basis for combining multiple analytical branches in one system (Table I).

Area	Approach
Stock screening	Rule-based filtering using market capitalization, P/E, P/B, revenue growth, dividend yield
Technical analysis	Price action and volume trends using Moving Averages, RSI, MACD
ML in stock analysis	Linear Regression, Decision Trees, Random Forest
Sentiment analysis	NLP of financial news, corporate reports, media text
Hill climbing	Local search moving to the neighbor with steepest score improvement

Table I: Approaches in the Literature

Screening and Technical Analysis: Traditional screeners narrow a large asset set using rule-based metrics. Technical analysis examines historical price and volume behavior with Moving Averages, the Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD) [7].

Machine Learning: Predictive and classification models such as Linear Regression, Decision Trees, and Random Forest [2] are used to identify non-linear patterns in financial data [5], [6].

Sentiment Analysis: Financial news, corporate reports, and media text carry information distinct from numerical indicators [3]. NLP classifies text as positive, neutral, or negative; sequence models such as LSTM [4] are related approaches.

Hill Climbing: Steepest-Ascent Hill Climbing is a local-search technique that considers neighboring states and moves to the one giving the steepest score improvement [1].

Research Gap: Users often rely on manual filters across disconnected tools, and explainable AI is not integrated into the same workflow. The proposed system combines screening, technical and fundamental analysis, and AI-driven explanations on one platform.

III. System Architecture

The system is a unified decision-support platform that receives stock tickers, screening preferences, and custom filters from the user. It follows a seven-layer pipeline (Fig. 1): User Layer, User Interface Layer (responsive web/mobile dashboard), Data Collection Layer, Data Processing Layer, Multi-Branch Analysis, AI/ML Engine, and Results and Reports.

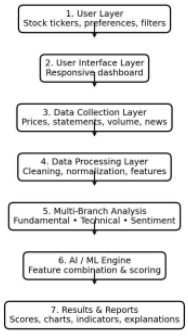


Fig. 1: Seven-layer architecture of the proposed system

A. Layer Description

The User Layer accepts stock tickers, screening preferences, and custom filter criteria. The User Interface Layer provides the dashboard through which users interact with the system and visualize results. The Data Collection Layer aggregates market prices, financial statements, trading volume, and news text, and the Data Processing Layer performs cleaning, normalization, and feature engineering. The Multi-Branch Analysis layer runs fundamental, technical, and sentiment analysis in parallel. The AI/ML Engine combines

features, performs pattern analysis, and generates scores and rankings. Finally, the Results and Reports Layer presents scores, comparative charts, indicators, and AI-generated explanations.

Stage	Main responsibility
Input	Stock tickers, screening preferences, custom filters
Interface	Responsive web/mobile dashboard
Collection	Market prices, financial statements, volume, news text
Processing	Cleaning, normalization, feature engineering
Analysis	Fundamental, technical, sentiment branches
AI/ML	Feature combination, pattern analysis, scoring/ranking
Output	Scores, charts, indicators, AI explanations

Table II: Seven-Stage Processing Pipeline

B. Data Collection, Processing, and Analysis

Data Collection and Processing: The collection layer aggregates market prices, financial statements, trading volume, and news text. The processing layer performs cleaning, normalization, and feature engineering so the data can be combined by the analytical engine.

Multi-Branch Analysis: Three branches run in parallel: fundamental analysis of company financials, technical analysis of price and volume history, and NLP-based sentiment analysis of financial text. Their outputs are combined by the AI/ML engine into a multi-factor representation rather than a single signal.

IV. Modules

A. Fundamental Analysis Module

This module evaluates financial ratios, earnings, and balance-sheet metrics to gauge underlying financial health. Its outputs are processed features, not a standalone decision; they contribute to the composite weighted score alongside technical and sentiment features.

Component	Role
Financial ratios	Quantitative measures for evaluating the company
Earnings	Important part of company-level financial evaluation
Balance-sheet metrics	Information for assessing underlying financial health
Fundamental features	Inputs to the combined AI/ML scoring process

Table III: Fundamental Analysis Components

B. Technical Analysis Module

This module tracks Moving Averages, RSI, MACD, and price/volume dynamics to evaluate historical trading patterns. Historical data is cleaned, normalized, and transformed into features; the dashboard can display key indicators alongside stock scores.

Indicator / Input	Purpose
Moving Averages	Track historical price trends
RSI	Technical indicator for evaluating historical patterns
MACD	Technical indicator for evaluating historical patterns
Price and volume dynamics	Historical market behavior for analysis

Table IV: Technical Analysis Indicators

C. Sentiment Analysis Module

Applying NLP to financial news, corporate reports, and media text, this module classifies information as positive, neutral, or negative. The resulting sentiment features add a textual perspective to the numerical branches and contribute to the plain-language explanation of each score.

The module takes three kinds of input—financial news, corporate reports, and media text—and converts them into sentiment information in three defined classes: positive, neutral, and negative. This sentiment information is passed to the AI/ML engine for feature combination and stock scoring.

D. AI/ML Engine Module

The engine combines features from all three branches using Random Forest, RNN/sentiment models, classification, and pattern analysis, and produces a composite weighted score for each stock. Scores are converted into rankings, and the system generates plain-language summaries explaining why a stock received its score.

Random Forest is listed as an analytical modeling approach, RNN/sentiment models for sentiment-related modeling, classification for analytical categorization, and pattern analysis for identifying patterns in the combined feature space. Hill climbing optimizes the scoring weights.

E. Steepest-Ascent Hill Climbing

A state is a configuration of scoring weights; neighboring states are alternative weight configurations. The algorithm proceeds as follows:

1) Start with an initial scoring-weight configuration.
2) Consider neighboring weight configurations.
3) Evaluate the score of each neighbor.
4) Select the neighbor with the steepest improvement.
5) Move to that state.
6) Repeat until no neighbor improves the score.

Within the architecture, hill climbing sits in the AI/ML engine. It does not collect data or perform the individual analyses; it optimizes the weights used to combine analytical features.

V. Implementation and Results

A. Experimental Setup

The system is designed as a responsive web/mobile dashboard backed by a multi-layer analytical pipeline (Table V).

Element	Description
Interface	Responsive web/mobile dashboard
Inputs	Tickers, preferences, custom filters
Data	Prices, statements, volume, news text
Processing	Cleaning, normalization, feature engineering
Analysis	Fundamental, technical, sentiment
AI/ML	Feature combination, scoring/ranking, optimization
Output	Scores, rankings, charts, indicators, explanations

Table V: Experimental Setup

In operation, a user supplies tickers and preferences; the system collects and processes numerical and textual data, runs the three branches in parallel, and combines their features. The AI/ML engine generates a composite score and ranking, and the dashboard presents the result (Fig. 2).

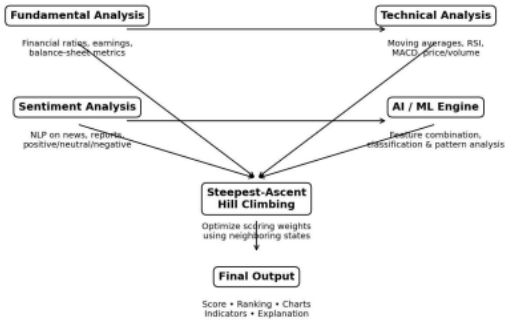


Fig. 2: Conceptual flow of the analysis modules and final output

Fundamental analysis contributes financial-health information, technical analysis contributes historical market-pattern information, and sentiment analysis contributes classified textual information. These branches feed the AI/ML engine, where features are combined and scoring weights are optimized. The final stage presents stock scores, rankings, charts, indicators, and AI-generated explanations.

B. Results

The system outputs are: (i) a **composite weighted score and ranking** for easy comparison of stocks; (ii) **dashboard charts** visualizing price trends, technical indicators, and multi-factor scores; (iii) **indicators**, with key technical and fundamental values shown alongside each result; and (iv) **AI-generated explanations**, plain-language summaries of why a stock received its score. Together these form a unified view of the complete result.

Output	Description
Score / Ranking	Composite weighted score and comparative ranking
Dashboard charts	Visual comparison of trends, indicators, multi-factor scores
Indicators	Key technical and fundamental indicators
AI explanations	Plain-language reasoning associated with the score
User dashboard	Unified visualization of the complete result

Table VI: System Outputs

C. Advantages, Limitations, and Applications

Advantages: unifies multiple analysis types, reduces manual screening across disconnected tools, combines numerical and textual information, optimizes scoring weights, and provides rankings, visualizations, and

explanations through a responsive dashboard.

Limitations: the system is a decision-support tool and does not guarantee returns; output quality depends on the quality and completeness of collected data; hill climbing is a local search based on neighboring-state improvement rather than an exhaustive global search; and no measured accuracy or financial-return benchmarks are reported.

Applications: stock screening and candidate comparison, multi-factor equity analysis, educational exploration of AI in financial technology, and consolidated decision support.

VI. Discussion

A. Why a Unified, Multi-Factor Approach

Each analytical branch captures a different view of a stock. Fundamental analysis reflects company-level financial health, technical analysis reflects historical price and volume behavior, and sentiment analysis reflects the information carried by financial text. Relying on any single signal gives an incomplete picture; combining the three in one engine yields a multi-factor representation, and placing them in one interface removes the need to move between disconnected tools.

B. Role of Weight Optimization

A composite score depends on how strongly each feature is weighted. Rather than fixing the weights by hand, the system treats each weight configuration as a state and uses Steepest-Ascent Hill Climbing to move toward configurations that improve the score, with the aim of favoring strong-performing stock rankings. Because the method evaluates neighboring configurations and stops when none improves the score, it is simple to reason about and well suited to iterative refinement of weights.

C. Explainability

Numerical scores alone can be difficult for beginners to interpret. The system therefore attaches a plain-language summary to each score, drawing on all three analytical inputs, so that users can see why a stock was ranked as it was and compare candidates more easily. Surfacing key indicators next to each score supports the same goal.

D. Practical Considerations

The quality of the output depends directly on the quality and completeness of the collected data, which is why cleaning, normalization, and feature engineering form a dedicated layer before analysis. Since hill climbing is a local search, it improves weights through neighboring-state moves rather than exhaustively searching all configurations. These considerations, together with the absence of measured benchmarks, mean the system should be read as an analytical aid, not a predictor of returns.

VII. Conclusion and Future Work

A. Summary of Contributions

The main contributions of this work are: (1) a unified interface that brings fundamental, technical, and sentiment analysis together; (2) a seven-layer architecture covering the path from user input to reporting; (3) the use of Steepest-Ascent Hill Climbing to optimize stock-scoring weights; and (4) explainable output in the form of scores, rankings, charts, indicators, and plain-language explanations.

B. Conclusion

This paper proposed a unified decision-support system combining fundamental, technical, and sentiment analysis through a seven-layer architecture connecting user input, interface, data collection, processing, multi-branch analysis, AI/ML processing, and reporting. Steepest-Ascent Hill Climbing optimizes the stock-scoring weights, and the system produces composite scores, rankings, charts, indicators, and plain-language explanations, aiming to reduce manual screening effort and make multi-factor analysis more accessible. It offers structured insight and explainable analysis, not financial advice or guaranteed returns.

C. Future Work

Development can extend the unified workflow, improve the data-processing pipeline, enrich explainability, and evaluate the scoring approach using clearly defined experimental measures such as accuracy and return benchmarks. These are development directions, not results already established by this work.

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